

Model Architecture: Spatiotemporal iEEG Embedding

Input

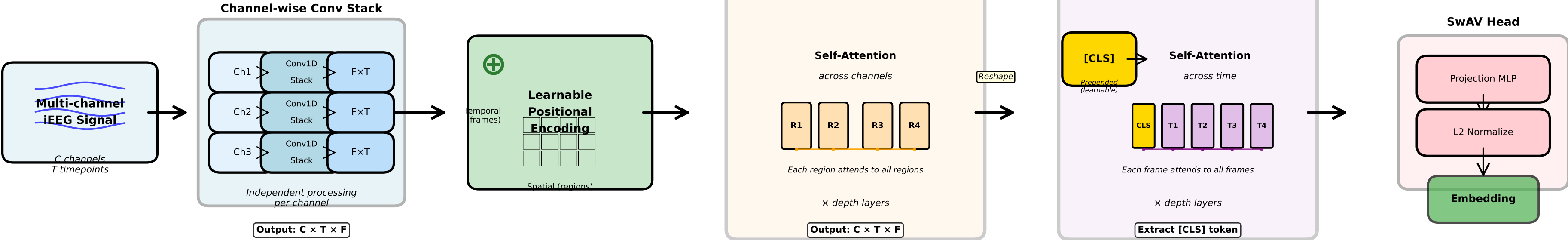
Feature
Extraction

Positional
Encoding

Spatial Processing

Temporal Processing

Output



Key Design Choices:

1. Per-channel convolution (independent feature extraction)
2. Learnable positional encoding (spatial + temporal)
3. Spatial attention: Regions communicate (which areas matter)
4. Temporal attention: Frames communicate (when events occur)
5. [CLS] token for global representation